Comprehensive Learning Portfolio & Progress Report

Introduction

This document serves as a comprehensive portfolio summarizing the active learning, practical exercises, and skill development undertaken to achieve my core educational goals. Documenting every granular step of every module was not feasible due to time constraints,because I am a medical student who learn a lot of slides in one period.. However, this report highlight my progress, and independent practices in different websites like in **freecodeacademy, and picoctf.** The report heavily supported by screenshot evidence—clearly demonstrates the substantial effort, consistency, and practical success achieved across three main disciplines: Cybersecurity (CTF challenges), Python Programming, and Professional English Communication.

1. Cybersecurity & Applied Cryptography (CTF Challenges)

This section demonstrates practical command-line skills, secure file handling, and basic cryptography techniques completed within the CyLab/picoCTF environment.

Challenge Execution: Obedient Cat, Netcat, and Password Cracking

Below is the terminal evidence showing the extraction of flags utilizing commands such as wget, cat, and nc, alongside the execution of Python scripts for hashing (MD5) to reverse engineer passwords.

A. Obedient Cat

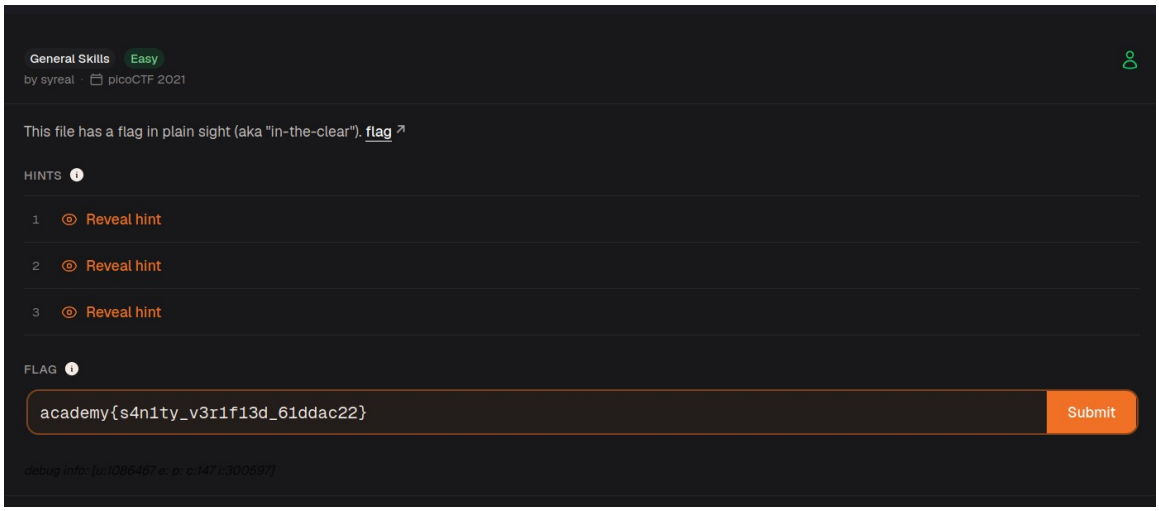
- I downloaded the file which contain the file and read it using “cat”

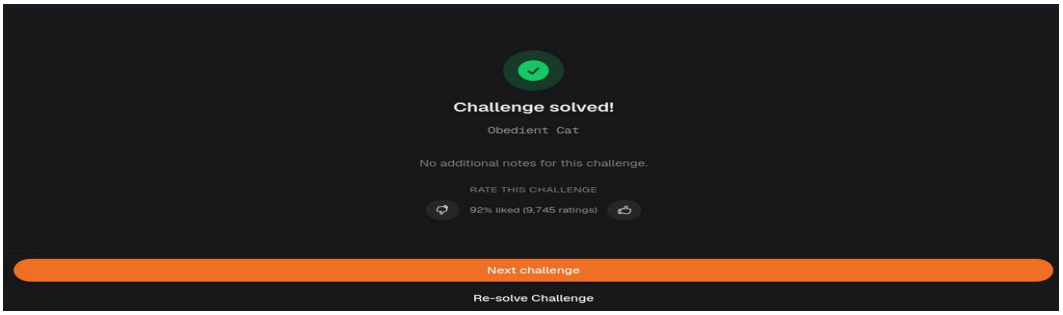
```
Sola25-academy@webshell:~$ wget "https://challenge-files.cylabacademy.net/library/7db7046a778dac83f8d537fe89f7cecaa851afb3c69e4c8c2464510aba9b47c2/flag"
--2026-10-26 19:12:35--  https://challenge-files.cylabacademy.net/library/7db7046a778dac83f8d537fe89f7cecaa851afb3c69e4c8c2464510aba9b47c2/flag
Resolving challenge-files.cylabacademy.net (challenge-files.cylabacademy.net)... 3.160.5.64, 3.160.5.18, 3.160.5.40, ...
Connecting to challenge-files.cylabacademy.net (challenge-files.cylabacademy.net)|3.160.5.64|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 34 [application/octet-stream]
Saving to: 'flag.1'

flag.1              100%[===============>]      34  --.-KB/s    in 0s

2026-10-26 19:12:35 (22.1 MB/s) - 'flag.1' saved [34/34]

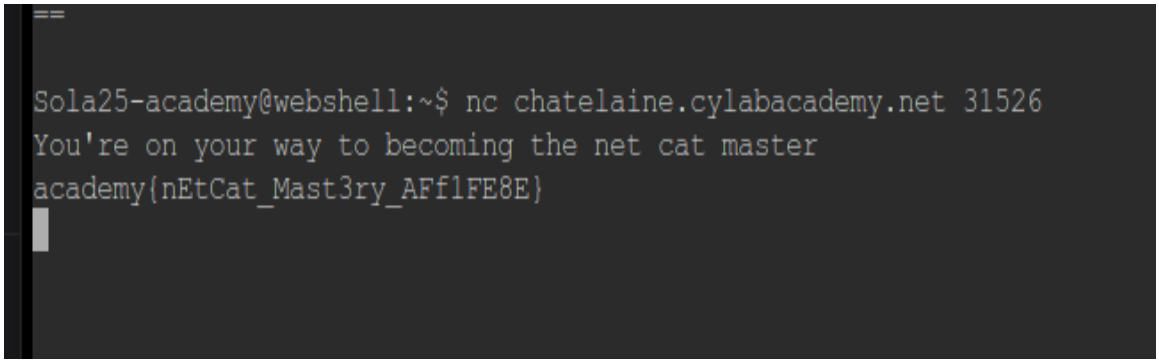
Sola25-academy@webshell:~$ ls
Addadshashanammu      flag                 level1.py            warm
Addadshashanammu.zip  flag.1               level2.flag.txt.enc  warm.1
README.txt            flag.txt.en          level2.py
ende.py               level1.flag.txt.enc  password.txt
Sola25-academy@webshell:~$ cat flag.1
academy{s4nlty_v3rlfl3d_61ddac22}
Sola25-academy@webshell:~$
```

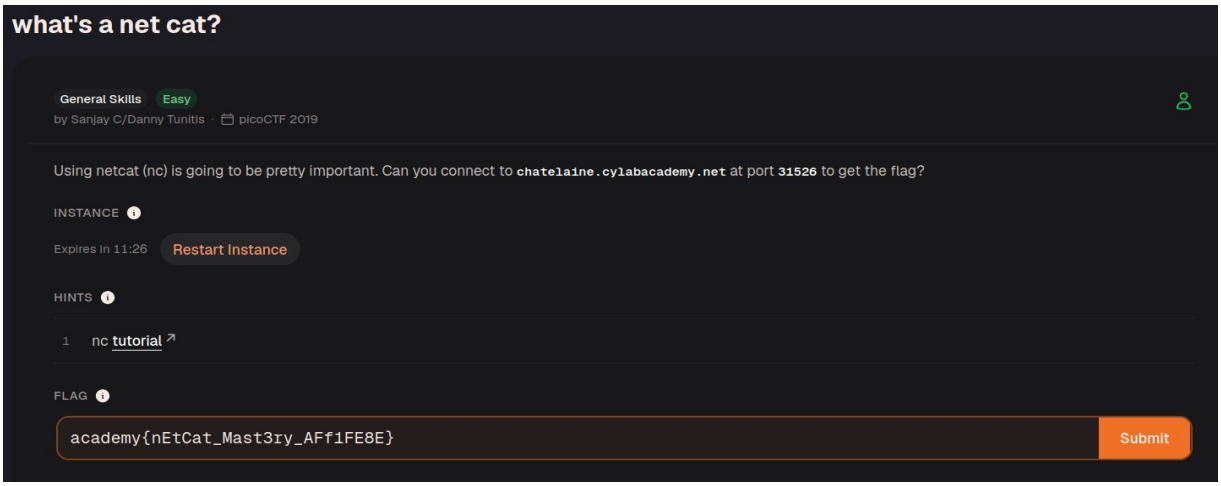




B. Netcat

- I launched Netcat to access the remote server's host name, and the port was 31526



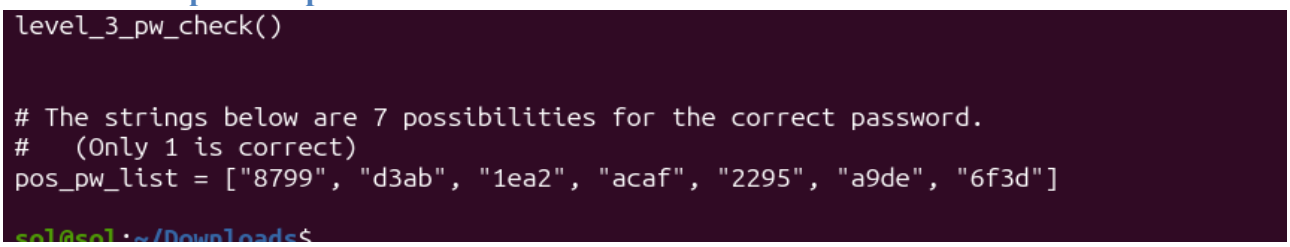


C. Password Cracking

1. First I downloaded the three essential files- Level3.py which contain the passwords, level3.flag.txt.enc, and level3.hash.bin



2. Seven possible passwords



3. I created a python code name pass.py to find a password that matches a given MD5 hash. To identify the correct password because the instruction state from seven passwords only one is correct.

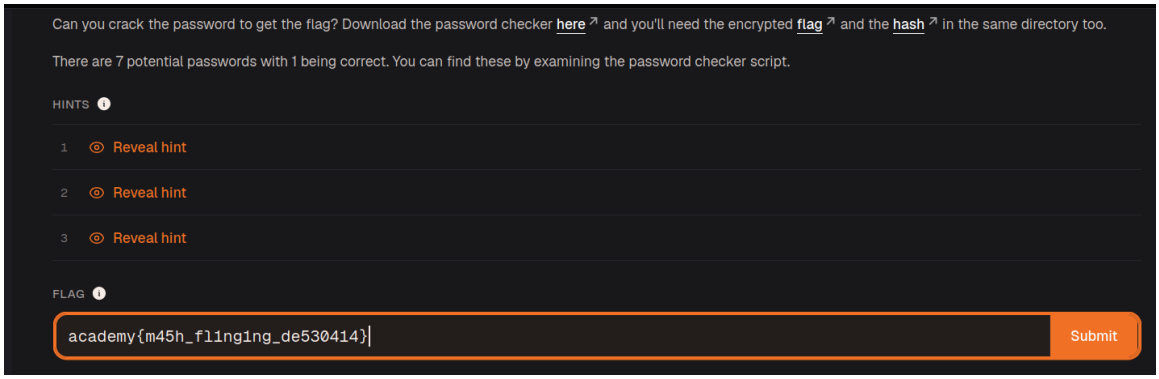
```
 GNU nano 8.7.1                                                pass.py
import hashlib

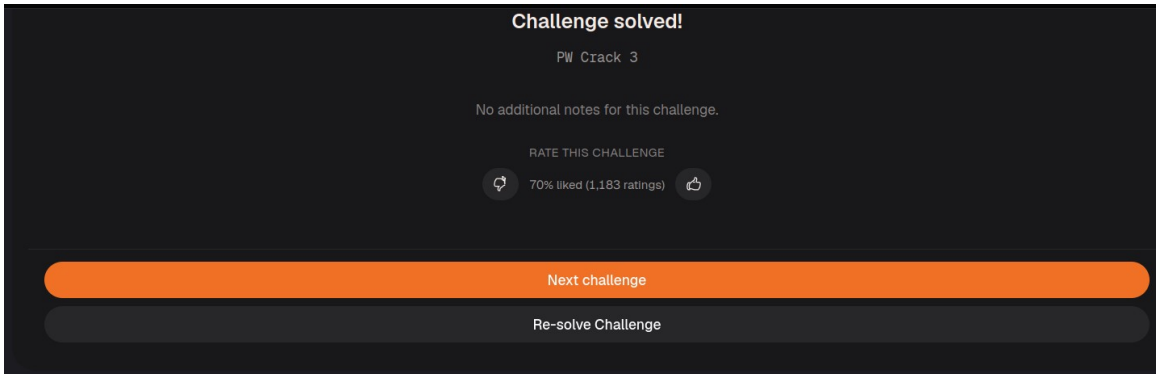
pos_pw_list = [
    "8799", "d3ab", "1ea2", "acaf", "2295", "a9de", "6f3d"
]

with open("level3.hash.bin","rb") as f:
   correct_hash = f.read()

for password in pos_pw_list:
    password_hash = hashlib.md5(password.encode()).digest()

    if password_hash == correct_hash:
       print("correct password:", password)
```





➢ picoCTF Platform Progression

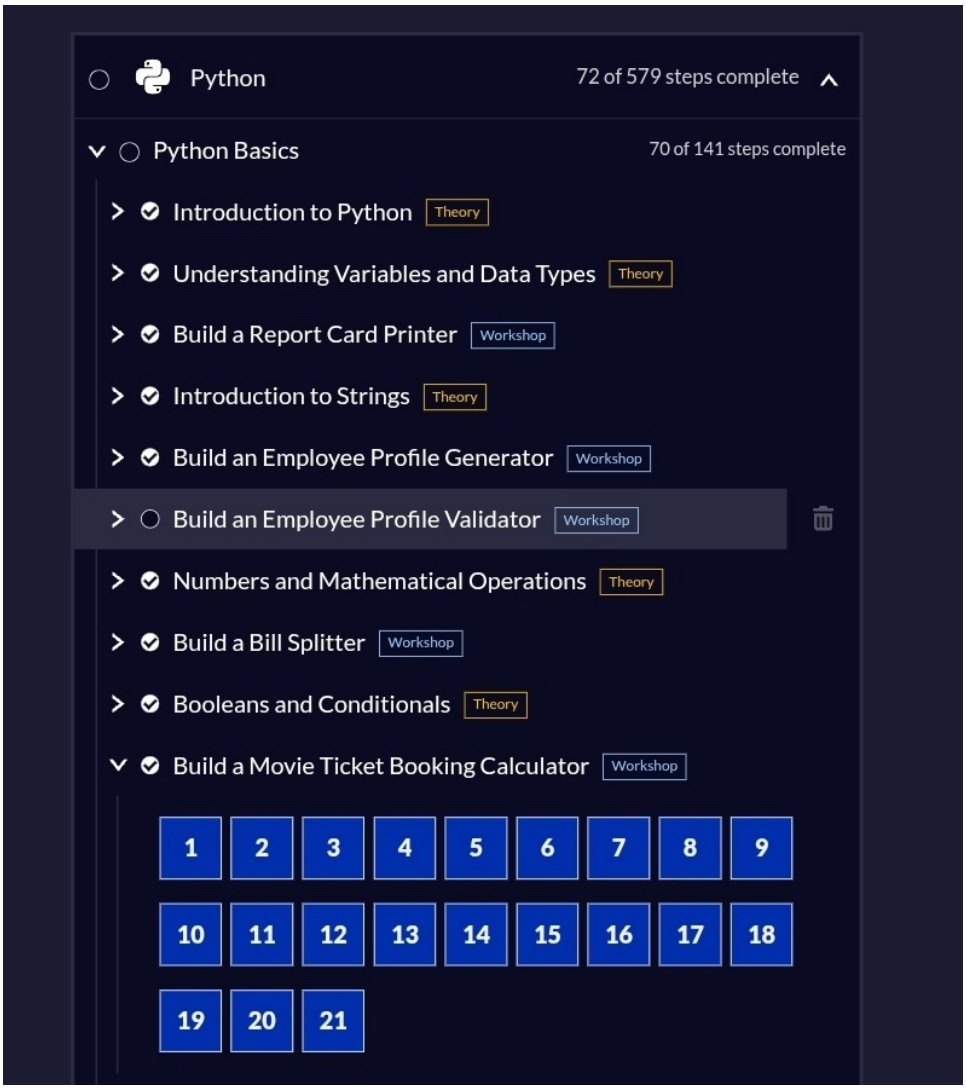
The following dashboard captures confirm my progression of numerous foundational CTF challenges (Sanity, CyberChef, Python, etc.):


CTF Onboarding
3/3 modules
What is a flag?
Reading
Traditional CTF Categ...
Reading
Approach
Reading
Section 1 (Sanity)
3/3 modules
Obedient Cat
Challenge
Super SSH
Challenge
what's a net cat?
Challenge


Section 2 (CyberChef)
4/4 modules
Mod 26
Challenge
Warmed Up
Challenge
2warm
Challenge
Bases
Challenge
Section 3
6/6 modules
Wave a flag
Challenge
Tab, Tab, Attack
Challenge
Insp3ctOr
Challenge
strings it
Challenge
First Grep
Challenge
where are the robots
Challenge

2. Python Programming Fundamentals

Consistent progress has been made in formal Python instruction, completing 72 out of 579 steps in the curriculum. The learning approach balances theoretical knowledge with hands-on workshops, covering variables, strings, booleans, and functional scripts like the Report Card Printer and Bill Splitter.



3. Professional English Language Practice

To complement technical skills, substantial effort has been dedicated to improving professional and workplace English communication. Completed modules include dialogues for formal introductions, office networking, and passing the 'First Day Conversations at Work Quiz'.

Learn Greetings in your First Day at the Office

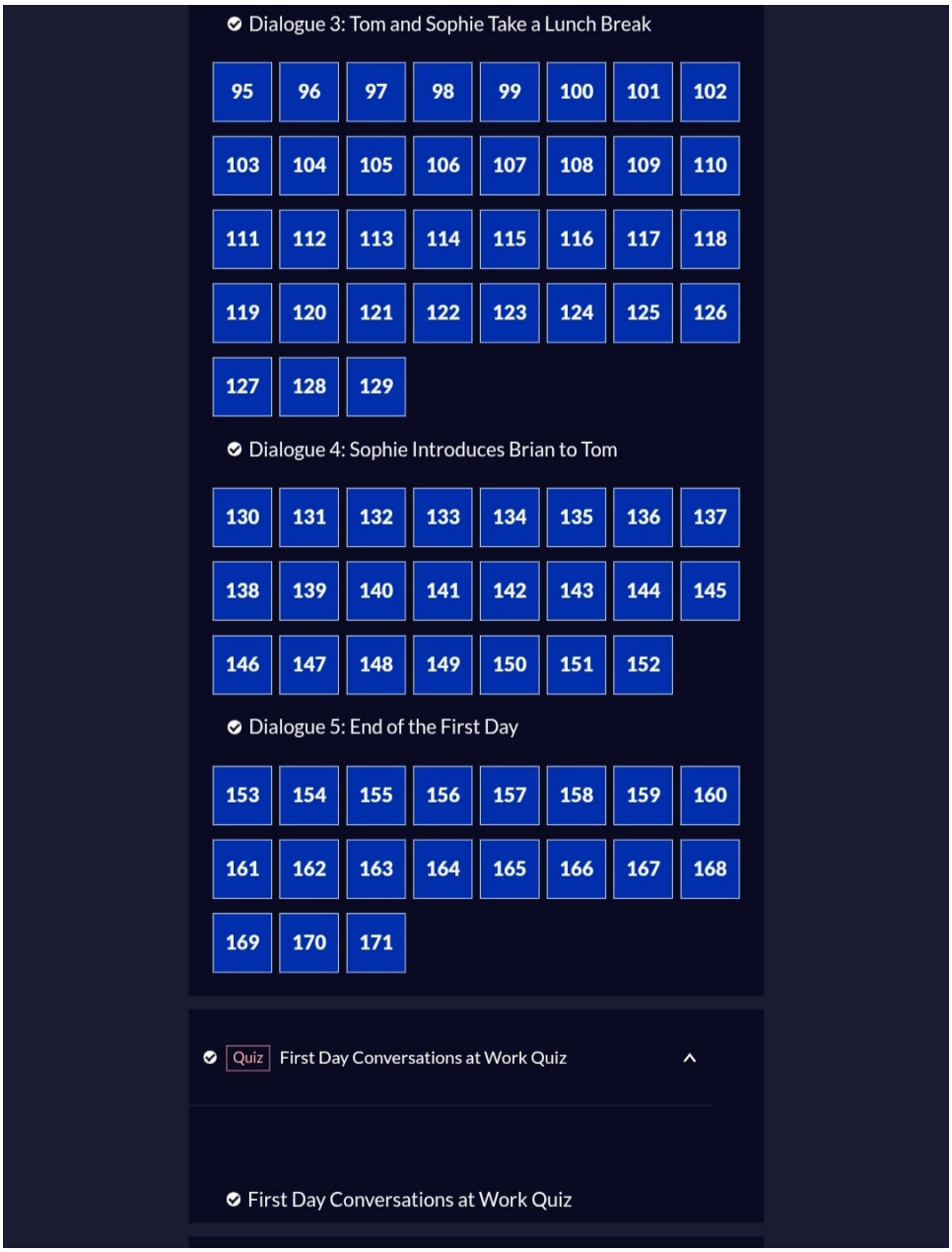
In this first course, you'll learn common expressions for situations you may encounter on your first day at work. You'll learn about introductions, getting to know people, asking for lunch recommendations, and getting an access card from security.

Dialogue 1: Maria Introduces Herself to Tom

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33							

Dialogue 2: Tom Meets the Coworker Next to Him

34	35	36	37	38	39	40	41
42	43	44	45	46	47	48	49
50	51	52	53	54	55	56	57
58	59	60	61	62	63	64	65
66	67	68	69	70	71	72	73
74	75	76	77	78	79	80	81



Conclusion

This portfolio highlights a multi-disciplinary commitment to professional growth. The combination of rigorous cybersecurity lab work, foundational software development, and English communication represents a holistic approach to achieving my career objectives.